

# Test Design Techniques — QA Portfolio

## Project Overview

This document demonstrates my understanding of three software testing techniques: equivalence partitioning, edge-case testing, and boundary value analysis. The practical checkout tests were performed on the Swag Labs demo application. Boundary value analysis is documented separately as a hypothetical exercise.

## 1. Equivalence Partitioning

**Objective:** Divide input data into groups expected to behave similarly and select representative test cases from each group.

Checkout form tests were designed to check valid information and missing required fields.

| Test Case | Scenario                      | Reported Outcome                               |
|-----------|-------------------------------|------------------------------------------------|
| TC06      | All checkout fields completed | Checkout overview opened                       |
| TC07      | All fields empty              | Validation error; remained on information page |
| TC08      | First Name missing            | Validation error; remained on information page |
| TC09      | Last Name missing             | Validation error; remained on information page |
| TC10      | Postal Code missing           | Validation error; remained on information page |

All five tests were reported as matching their expected outcomes.

## 2. Edge-Case Testing

**Test Case:** TC11 — Submit checkout with spaces-only First Name

**Test data:**

First Name: Three spaces

Last Name: Smith

Postal Code: 12345

**Expected behaviour:** The application should reject a First Name containing only whitespace.

**Reported actual result:** The application rejected the submission and displayed a validation error.

**Conclusion:** The observed behaviour matched the intended test expectation. The exact error text was not recorded. No defect is claimed without comparing the behaviour against documented requirements.

### 3. Boundary Value Analysis

**Objective:** Test values at and around the minimum and maximum limits of a specified input range.

**Hypothetical requirement:** A field accepts between 1 and 10 characters, inclusive.

| Test ID | Input         | Length | Expected Result |
|---------|---------------|--------|-----------------|
| BVA01   | 0 characters  |        | Reject          |
| BVA02   | 1 character   |        | Accept          |
| BVA03   | 2 characters  |        | Accept          |
| BVA04   | 9 characters  |        | Accept          |
| BVA05   | 10 characters |        | Accept          |
| BVA06   | 11 characters |        | Reject          |

**Analysis:** The minimum and maximum valid boundaries are 1 and 10 characters. Values 0 and 11 fall outside the permitted range. Values 2 and 9 are immediately inside the boundaries.

These cases illustrate test design and are not actual Swag Labs execution results. The example requirement has not been verified against Swag Labs specifications.

### 4. Key Learnings

Equivalence partitioning helps reduce redundant test cases by selecting representative inputs.

Edge-case testing checks unusual inputs such as whitespace-only values.

Boundary value analysis targets errors near input limits.

Expected and actual results must be recorded separately.

Test failures and defects should only be reported when observed behaviour contradicts a defined requirement.

## Tools and Environment

Application: Swag Labs demo website

Test management: Qase

Repository and portfolio: GitHub

Test types: Functional, negative, input-validation, edge-case and test-design exercises